PHARMACY SYSTEM DATABASE

Project Documentation

This project is a simple pharmacy database designed to store medicine information, customer details, and sales transactions. It uses PostgreSQL and SQL commands to create and manage the database.

1. Project Objectives

• Store details about medicines, including category, price, and available stock.
• Store customer contact information.
• Record sales and connect each sale to a customer and a medicine.
• Retrieve related information using joins and update data when needed.

2. Database Tables

Table	Purpose	Main columns
medicines	Stores medicine details	medicine_id (PK), medicine_name, medicine_category, price, stock_quantity
customers	Stores customer details	customer_id (PK), customer_name, phone_number (unique), city, email
sales	Stores each sale	sale_id (PK), customer_id (FK), medicine_id (FK), quantity_sold, sale_date

3. Relationships

• customers to sales: customer_id in sales references customer_id in customers. One customer can have many sales, while each sale is linked to one customer.
• medicines to sales: medicine_id in sales references medicine_id in medicines. One medicine can appear in many sales, while each sale refers to one medicine.

4. SQL Operations Used

The SQL script demonstrates database and table creation, primary keys, foreign keys, required fields (NOT NULL), a unique phone number, inserting records, selecting data, joining tables, updating records, and altering table structures by renaming a column, changing a data type, and adding an email column.

5. Conclusion

The Pharmacy System database organizes medicine, customer, and sales data in related tables. Primary keys identify records, and foreign keys help maintain relationships between tables. SQL queries can be used to view, add, and update the stored information.

Note: Review the table and column names in this document against your final SQL script before submission.